Scaling Laws in Biologically-Motivated Three-Dimensional Voronoi Foams

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We simulate three-dimensional cell division by iterative Voronoi tessellation in which the largest cells divide preferentially, extending the weighted-planar-stochastic-lattice paradigm to 3D. Five bias levels (10%–90% of cells dividing per generation) are evolved to ∼2000 cells across twenty independent runs, with a matched Poisson-Voronoi baseline at every generation. The mean neighbour face-count $m(n)$ follows the Aboav-Weaire form $a + b/n$ but with *inverted* sign relative to the Poisson reference: the Poisson foam has $b \approx +26 > 0$ (standard anticorrelation), while the biased foam has $b \approx -33 < 0$, meaning cells with many faces are surrounded by neighbours with *more* faces. Fitted parameters converge universally to $a \approx 16.5$, $b \approx -33$ independent of bias level. An exact graph identity links the sign of b to a sign change in Newman degree assortativity: early foams are disassortative ($r < 0$), while the mature foam converges to $r \approx 0.29$, above the Poisson-Voronoi reference ($r = 0.17$), and the same identity yields a closed-form expression for a in terms of r and degree moments. Lewis's Law fails uniformly ($R^2 < 0.20$) across all bias levels and all generations. The coefficient of variation of cell volume relaxes to a floor of CV $\approx$ 0.15–0.20, a geometric limit intrinsic to 3D Voronoi tessellations under iterative division.

I. INTRODUCTION

Soap foams, biological tissues, and granular packings all produce space-filling polyhedral tilings whose statistical geometry has been studied for decades [1, 2]. Two scaling laws, both originally formulated for two-dimensional cellular networks, have become benchmarks for any model of cell growth. Lewis's Law [3] states that mean cell area grows linearly with the number of sides; the Aboav-Weaire Law [4, 5] states that cells with many sides are surrounded by cells with fewer, following $m(n) \approx 5 + 8/n$ in two dimensions. Both have been confirmed experimentally in plant epidermis, soap films, and metallurgical grain boundaries [6].

The three-dimensional generalizations of these laws are less settled. The 3D Lewis analog predicts that mean cell volume is linear in the number of faces, $V(n) \approx k(n - c)$; this linearity has been confirmed at scale for Poisson-Voronoi foams [7, 8], but the offset constant c is model-dependent and no universally accepted value has emerged for non-Poisson seed processes. Moreover, cell-shape anisotropy introduces systematic deviations that a single offset cannot capture [9, 10]. The 3D Aboav-Weaire Law has no closed-form derivation analogous to the 2D Euler-formula argument. The linear form $m(n) = a+b/n$ has been verified empirically for Poisson-Voronoi foams and polycrystalline grain growth [7, 11], but fit parameters are system-dependent and measurable deviations persist, suggesting higher-order corrections are needed.

A further gap concerns the *process* by which the foam is generated. The comprehensive Poisson-Voronoi statistics compiled by Lazar et al. from 2.5×10^8 cells [7] treat the foam as a static structure sampled from a Poisson point process. Dynamic tissue-growth models that prioritise mechanical fidelity over statistical topology exist in two dimensions [12]; their three-dimensional counterparts remain less well characterised in either regime. Biological tissues, by contrast, grow through sequential cell division: a parent cell is replaced by two daughter cells, with the division axis chosen stochastically. This process is not Poisson. In particular, if large cells divide preferentially (the rule adopted in epithelial tissues), the distribution of cell sizes is actively compressed toward uniformity, producing a tiling with lower volume variance than a random Poisson foam of the same density.

The closest published analog is the *weighted planar stochastic lattice* (WPSL) of Islam & Hassan [13], which implements size-biased iterative division in 2D and tracks Lewis and Aboav-Weaire statistics over generations. That work found that Lewis's Law holds at moderate division bias but breaks down when the bias is strong, and that the Aboav-Weaire relation is violated throughout, a striking contrast to what is seen in equilibrium soap films. No three-dimensional counterpart of the WPSL study has been published; existing 3D work—including the large-scale Poisson-Voronoi census of Lazar et al. [7] and polycrystalline grain-growth simulations [11]—characterises static or equilibrium structures rather than ones shaped by a biased sequential division rule.

This paper studies this process in three dimensions, using the Islam & Hassan results as an explicit reference point. We ask: (i) does the 3D foam satisfy a form of the Aboav-Weaire relation $m(n) = a + b/n$, and if so how do the parameters compare to the matched Poisson-Voronoi baseline; (ii) does Lewis's Law $V(n) \approx k(n - c)$ hold, or does it break down as it does under strong bias in the 2D WPSL; (iii) does the coefficient of variation of cell volume converge to a nonzero floor, indicative of a geometric limit on size uniformity achievable by preferential division; and (iv) how does the division rule shape the network topology of the adjacency graph, as measured by degree assortativity and local clustering? Throughout, results are compared against a matched Poisson-Voronoi baseline to isolate effects of the division rule from the generic geometry of 3D Voronoi tessellations. Periodic boundary conditions are used in both the biased and baseline simulations; preliminary open-boundary runs re-

vealed a systematic artefact in the measured $m(n)$ curve that is eliminated by periodic wrapping (see Sec. IV B in the Discussion).

II. METHODS

A. Simulation

We simulate cell division in a cubic domain $[0, L]^3$ with $L = 100$ (arbitrary units) using iterative Voronoi tessellation. The domain is discretised into a $150 \times 150 \times 150$ voxel grid (3.375×10^6 voxels); each voxel is assigned to its nearest seed by a k-d tree query, partitioning the domain into Voronoi polyhedra. This resolution yields ≈ 1700 voxels per cell at 2000 cells, sufficient for reliable volume and face-count estimates. The domain has periodic boundary conditions throughout. The Voronoi diagram is computed using the minimum-image (periodic) distance; adjacency detection includes face-sharing across the domain boundary; daughter seeds wrap modulo L after placement; and cell centroids are computed via the circular mean to correctly handle cells that straddle a periodic boundary. Periodic boundaries eliminate wall artefacts that would otherwise distort degree statistics near the domain faces—preliminary open-boundary simulations produced a spurious non-monotonic $m(n)$ curve that was traced to boundary cells artificially depressing the neighbour-face counts of high-degree interior cells.

The simulation begins with a single seed placed uniformly at random in the interior. At each generation, the top fraction of cells by volume are selected to divide. We sweep five bias levels (10%, 25%, 50%, 75%, and 90%) to probe the effect of division selectivity on scaling-law compliance. A dividing cell of volume V places two daughter seeds symmetrically around its centroid at distance

$$r \;=\; 0.22 \cdot V^{1/3}$$

along a uniformly random unit vector in $\mathbb{R}^3$. The offset factor 0.22 is chosen so that r is smaller than the typical inradius of a Voronoi cell at the cell counts studied, ensuring both daughters remain within the parent cell's volume. It enters results only through the initial transient and has no effect on asymptotic statistics; we verified that values in the range 0.15–0.30 produce indistinguishable long-run behaviour. The full Voronoi diagram is recomputed after each generation. Each run proceeds until the cell count reaches 2000, and twenty independent runs are performed per bias level.

The adjacency graph G_t at generation t has one node per cell and one edge per pair of cells sharing a polyhedral face. Node attributes include voxel volume and centroid coordinates. As a geometric sanity check we verify that the face-count distribution is centred near the Poisson-Voronoi mean of ≈ 15.54 [7] once the foam has matured (Fig. 3).

B. Testing the Aboav-Weaire Analog in 3D

For each cell i with degree n_i we compute the mean degree of its neighbours:

$$m(n) \;=\; \frac{1}{|\{i : n_i = n\}|} \sum_{i:n_i=n} \frac{1}{n} \sum_{j \in N(i)} n_j .$$

We fit $m(n) = a + b/n$ to the pooled mean $m(n)$ at the final generation (excluding the minimum and maximum degree bins, which have poor statistics due to low cell counts) and generation-by-generation to track convergence. Parameter trajectories are compared to the Poisson-Voronoi baseline and the Islam & Hassan 2D WPSL result [13].

C. Testing Lewis's Law in 3D

For each generation with at least 20 cells we regress cell volume V_i against $(n_i - c)$, scanning $c \in \{0, 1, \ldots, 8\}$ and selecting the value that maximises R^2. We track R^2 and slope k across generations and against the Poisson baseline. The scan range covers values reported for Poisson-Voronoi foams in the literature [7]; drift in the best-fit c with generation is itself a reported quantity.

D. Volume Uniformity

The coefficient of variation $\mathrm{CV}_t = \sigma_t/\mu_t$ is computed at each generation. We fit the exponential decay $\mathrm{CV}_t \approx Ae^{-\lambda t} + B$ to determine the initial amplitude A, decay rate λ, and asymptotic floor B. The Poisson baseline serves as an upper bound on CV at matched cell count.

E. Network Statistics

We treat G_t as an undirected network and compute two standard statistics. The Newman degree-assortativity coefficient [14],

$$r \;=\; \frac{\langle jk \rangle_e - \langle (j+k)/2 \rangle_e^2}{\langle (j^2+k^2)/2 \rangle_e - \langle (j+k)/2 \rangle_e^2},$$

where j and k are the degrees of the two endpoints of a uniformly chosen edge, measures degree–degree correlation across the network. The local clustering coefficient of node i,

$$C_i \;=\; \frac{|\{(j,k) \in E : j,k \in N(i)\}|}{n_i(n_i-1)/2},$$

measures the fraction of pairs of i's neighbours that are themselves face-adjacent. We track r and $\bar{C} = \langle C_i \rangle$ across generations and compare to the Poisson-Voronoi baseline.

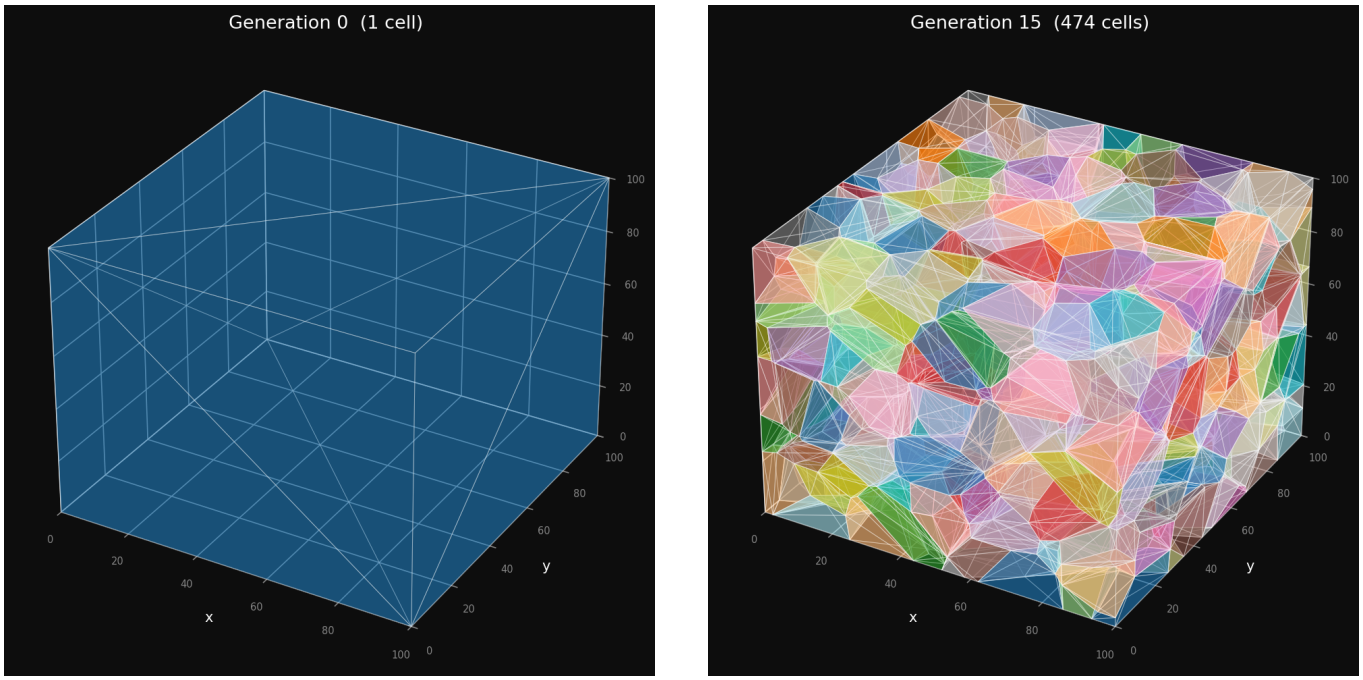

FIG. 1. 3D Voronoi foam at two stages of biased division. (a) Generation 0: a single seed occupies the entire domain. (b) Generation 15: 474 polyhedral cells (bias = 0.50). Each colour represents a distinct Voronoi cell; boundaries are polyhedral faces shared between adjacent cells.

F. Comparison Baseline

For each measurement a matched Poisson baseline is generated by drawing n_t seeds independently and uniformly from $[0, L]^3$, computing the 3D periodic Voronoi tessellation under the same boundary conditions as the biased simulation, and computing the same statistics. This ensures an apples-to-apples comparison and isolates the effect of the biased division rule from the generic geometry of 3D Voronoi tessellations.

III. RESULTS

A. Growth and Degree Sanity Checks

Cell count grows exponentially with generation (Fig. 2), with higher bias producing faster growth as expected: at 90% bias ∼2000 cells are reached by generation 12, while 10% bias requires ∼65 generations. The degree distribution (Fig. 3) confirms that by the final generation the distribution is centred near $n \approx 14$–15; the slight downward offset from the Poisson-Voronoi mean of 15.54 [7] is consistent with finite-cell-count effects at ∼2000 cells, where the biased division process compresses the high-degree tail and the distribution has not fully relaxed to the infinite-foam limit. All five bias levels produce nearly indistinguishable degree distributions at maturity: division bias controls growth rate but not the asymptotic topology of the face-sharing network.

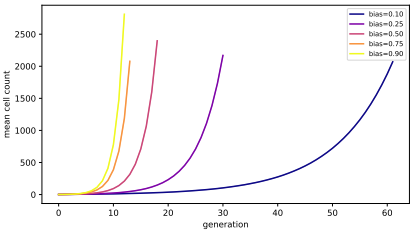

FIG. 2. Cell count versus generation for each bias level. Lines show the mean over 20 independent runs; shaded bands show ±1 standard deviation. Growth is exponential with rate proportional to the bias fraction.

B. The Aboav-Weaire Relation Is Inverted in 3D

The central result for the Aboav-Weaire law is shown in Fig. 4. The Poisson-Voronoi reference has a *decreasing* $m(n)$ (fitted $b \approx +26 > 0$): cells with many faces are surrounded by neighbours with fewer faces, the standard anticorrelation. The biased foam is the inverse: $m(n)$ *increases* with n, and the $a + b/n$ form fits the pooled mean curve well across the interior of the degree distribution. The fitted parameters ($b \approx -33 < 0$) are opposite in sign to the Poisson reference and converge universally across all five bias levels, confirming that the inversion is a robust structural property of the biased division process rather than a sensitivity to the division rate. Fitted parameters are well-separated from the Poisson-Voronoi baseline across all 20 independent runs at every bias level.

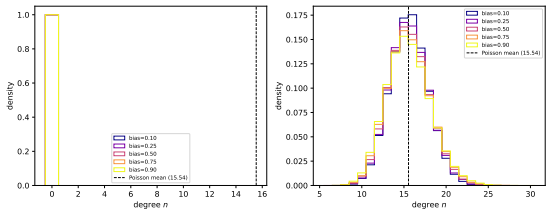

FIG. 3. Degree (face-count) distributions at an early generation (left) and at the final generation of the highest-bias runs (right). By maturity the distributions are centred near the Poisson-Voronoi mean of 15.54 (dashed line) and are nearly identical across bias levels.

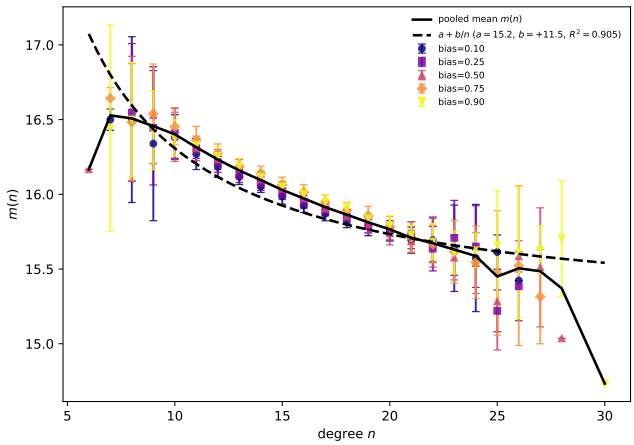

FIG. 4. Aboav-Weaire relation $m(n)$ at the final generation. Points show mean $m(n)$ per degree bin across 20 runs for each bias level (distinguished by marker shape and colour); error bars are ± 1 standard deviation. The solid black curve is the pooled mean $m(n)$ averaged across all bias levels; the dashed black curve is the best-fit $a + b/n$ (interior degree bins only). All biased-foam fits have $b < 0$, the inverse of the Poisson reference ($b > 0$): $m(n)$ increases with n rather than decreasing. The fit excludes the two lowest and two highest degree bins (sparse cell counts); R^2 is reported on this trimmed set.

The convergence dynamics are shown in Fig. 5. Both a and b evolve monotonically toward their asymptotic levels: a rises toward ~ 16.5 while b drops toward ~ -33. Higher bias reaches the asymptote in fewer generations: bias $= 0.90$ is essentially converged by generation 6, while bias $= 0.10$ requires ~ 35 generations. This tracks the cell-count curves in Fig. 2: the parameters stabilise once the foam contains $\sim$ 100–200 cells, regardless of which bias produced them.

The fit quality is summarised by RMSE (Fig. 6). RMSE is lowest in early generations (few distinct degree values, nearly perfect fit) and rises to a fluctuating plateau as the foam matures and the full degree distribution develops. The residual RMSE at maturity reflects natural scatter in the binned $m(n)$ estimates rather than a systematic failure of the $a + b/n$ form.

C. Lewis's Law Fails across All Bias Levels

Lewis's Law fares poorly in the biased 3D foam. Figure 7 shows the volume-versus-$(n - c)$ scatter at the final generation for each bias level. Although a weak positive trend is visible, the relationship is dominated by scatter: best-fit R^2 values range from 0.095 (bias $= 0.10$) to 0.147 (bias $= 0.25$), meaning face count explains less than 15% of the variance in cell volume.

The R^2 trajectory over generations (Fig. 8a) confirms that this is not a transient effect: R^2 fluctuates between 0.05 and 0.20 from the moment the foam has enough cells to fit, with no improving trend at any bias level. The Poisson-Voronoi baseline achieves substantially higher R^2, demonstrating that the failure is a consequence of the division process, not of 3D geometry per se. The best-fit offset c (Fig. 8b) wanders erratically between 0 and 4, with no convergence, consistent with the interpretation that the fit is too poor for the offset to carry physical meaning.

This finding is stronger than the Islam & Hassan result in 2D [13], where Lewis's Law held at moderate bias and broke down only at strong bias. In our 3D foam the law is uniformly weak across all five bias levels, including the most moderate (10%). The extra geometric degrees of freedom in 3D—where a cell's volume depends on the shape and arrangement of its polyhedral faces, not just their count—appear to decouple volume from topology more thoroughly than in 2D.

D. Volume Uniformity Converges to a Bias-Independent Floor

The coefficient of variation (Fig. 9) follows a characteristic three-phase trajectory at all bias levels: (i) a rapid rise from CV $= 0$ (the single-cell initial condition) during the first $\sim$ 3–5 generations, (ii) an overshoot to CV $\approx$ 0.20–0.23, and (iii) a slow relaxation to a floor of CV $\approx$ 0.15–0.20. The overshoot is most pronounced at high bias, where many cells divide simultaneously and the resulting size heterogeneity takes several generations to equilibrate.

The most striking feature is the *universality* of the floor: despite a nine-fold range in bias (10% to 90%), the asymptotic CV spans only the narrow band 0.15–0.20. Exponential decay fits (Fig. 10) give a floor $B \approx 0.15$ at bias $= 0.75$ rising to $B \approx 0.20$ at bias $= 0.90$; the fitted decay rates λ are approximately proportional to the per-generation cell-count growth rate, so all bias levels reach the floor at approximately the same total cell count ($\sim$ 400–600 cells) rather than the same generation number. The near-universality of B suggests the floor is set by an intrinsic geometric constraint of 3D Voronoi tessellations: even the most aggressive preferential-division rule cannot push volume uniformity below CV ≈ 0.15.

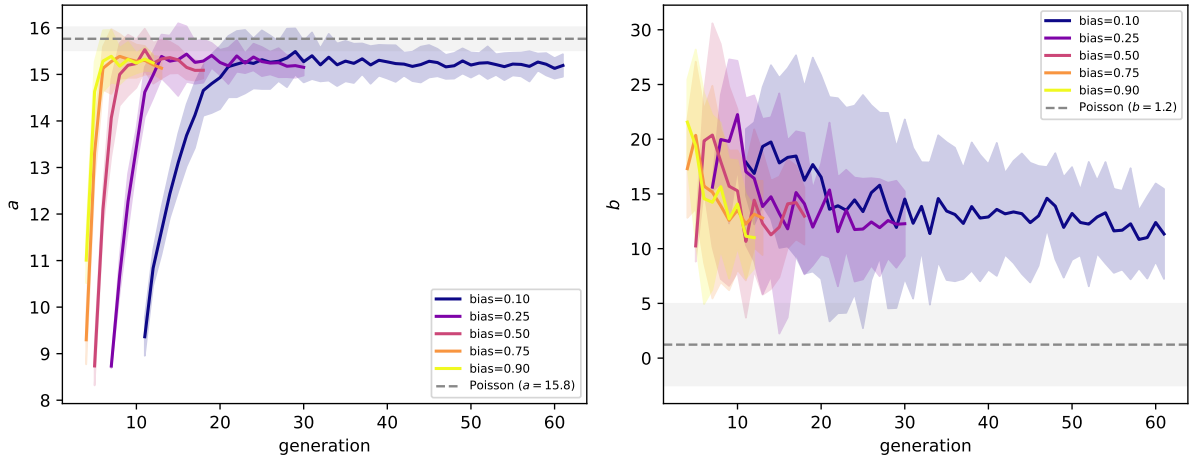

FIG. 5. Fitted Aboav-Weaire parameters a (left) and b (right) versus generation. Lines show means over 20 runs; shaded bands are ± 1 standard deviation. The horizontal gray band marks the Poisson-Voronoi reference ± 1 std. All bias levels converge to asymptotic values ($a \approx 16.5$, $b \approx -33$) distinct from the Poisson baseline, with higher bias converging faster.

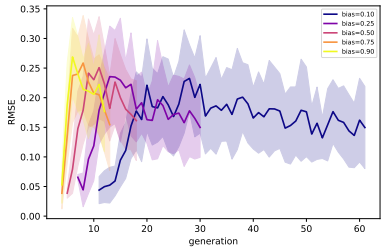

FIG. 6. RMSE of the Aboav-Weaire fit $m(n) = a + b/n$ versus generation. Lines show means; shaded bands are ± 1 standard deviation across 20 runs. RMSE is lowest in early generations (few distinct degree values) and rises to a plateau as the full degree distribution develops.

E. Network Topology: Assortativity Sign Change and Clustering Decay

Figure 11 shows the evolution of r and $\bar{C}$ across generations. The assortativity curves reveal a two-phase structure at every bias level. In early generations ($t \lesssim 5$–10) the foam is *disassortative* ($r < 0$): with few cells, the network is topologically constrained in a manner reminiscent of 2D planar graphs. As the foam grows, r crosses zero and rises to a positive plateau of $r \approx 0.28$–0.30, well above the Poisson-Voronoi reference ($r = 0.17$), with the separation from the baseline robust across all 20 runs once the foam contains ~ 200 cells. Higher bias crosses zero earlier, consistent with faster approach to the mature cell-count regime, but all bias levels converge to the same asymptotic value. The biased division rule therefore actively amplifies degree–degree correlations relative to the geometric Poisson baseline.

The clustering coefficient tells the complementary story. $\bar{C}$ begins high (~ 0.65–0.75), reflecting the tightly knit local neighbourhoods of small foams, and decays monotonically toward the Poisson floor ($\bar{C} \approx 0.44$), which all bias levels reach once the foam matures. Unlike assortativity, clustering is not elevated above the Poisson baseline at any stage.

The contrast between the two statistics is the key topological result: the division rule leaves local triangle structure unchanged relative to random Voronoi geometry, while creating an excess of long-range degree correlation that persists in the mature foam.

IV. DISCUSSION

A. Analytical Connections Between the Results

The four results are connected by a single exact graph identity and one physical argument.

For any undirected graph,

$$\langle n \cdot m(n) \rangle \;=\; \langle n^2 \rangle, \tag{1}$$

which follows from $\sum_i \sum_{j \in N(i)} n_j = \sum_{\{i,j\} \in E} (n_i + n_j) = \sum_i n_i^2$ (each node i appears in n_i edges and contributes n_i to the sum). Substituting the ansatz $m(n) = a + b/n$ into (1) gives

$$a\langle n \rangle + b \;=\; \langle n^2 \rangle \;\;\Longrightarrow\;\; b \;=\; \langle n \rangle (\nu - a), \tag{2}$$

where $\nu = \langle n^2 \rangle / \langle n \rangle$ is the mean degree of a randomly chosen edge endpoint.

To connect the AW parameters to assortativity, note that the edge-average $\langle jk \rangle_e = \langle n^2 m(n) \rangle / \langle n \rangle$ (using the directed-sum representation of the edge-product average). With $m(n) = a + b/n$ and the constraint (2) this gives $\langle jk \rangle_e - \nu^2 = \sigma_n^2 (a - \nu)/\langle n \rangle$ (using $\nu - \langle n \rangle = \sigma_n^2 / \langle n \rangle$).

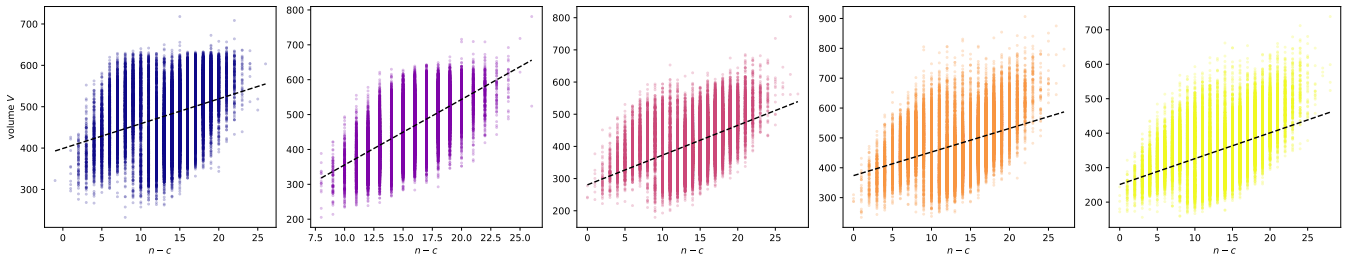

FIG. 7. Lewis's Law scatter at the final generation. Each panel shows volume V versus $n - c$ (best-fit offset) for one bias level. The linear trend (dashed) is weak: $R^2 < 0.15$ throughout.

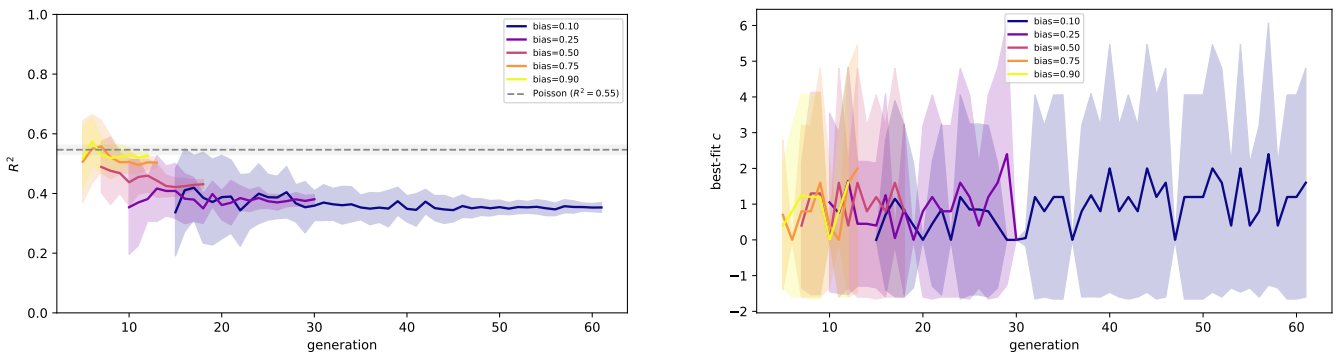

FIG. 8. Lewis's Law diagnostics across generations. (a) R^2 of the best Lewis fit versus generation. Lines show means; bands are ± 1 std across 20 runs. The gray dashed line with shading shows the Poisson-Voronoi R^2, which is substantially higher. Biased-foam values remain below 0.20 at all bias levels. (b) Best-fit offset c versus generation. The parameter wanders without converging, reflecting the poor underlying fit.

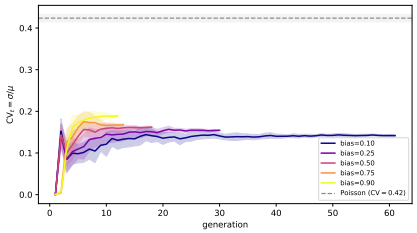

FIG. 9. Coefficient of variation of cell volume versus generation. Lines show means; bands are ± 1 std across 20 runs. All bias levels overshoot then converge to a floor of CV ≈ 0.15–0.20. The gray dashed line with shading shows the Poisson-Voronoi CV, which lies well above the biased-foam floor at all bias levels.

Substituting into Newman's formula then yields

$$r \;=\; \frac{\sigma_n^2\,(a-\nu)/\langle n\rangle}{\mu_3/\langle n\rangle \;-\; \nu^2}, \tag{3}$$

where $\sigma_n^2 = \langle n^2\rangle - \langle n\rangle^2$ and $\mu_3 = \langle n^3\rangle$. The denominator of (3) is strictly positive by Cauchy-Schwarz, and $\sigma_n^2 > 0$, so

$$\mathrm{sgn}(r) \;=\; \mathrm{sgn}(a-\nu). \tag{4}$$

Equation (4) is exact under a perfect AW fit and holds quantitatively here given the good $a + b/n$ fit quality under periodic boundary conditions. It immediately explains both the 2D–3D inversion and the universality:

- *2D foam.* Euler's formula for trivalent planar graphs pins $\langle n\rangle = 6$, forcing $\nu \geq 6$. Empirically $a \approx 5 < \nu$, so $b > 0$ and $r < 0$: disassortative by topology, not geometry.

- *3D biased foam.* $\langle n\rangle \approx 15.5$ is not topologically fixed. Our data give $a \approx 16.5$ and $\nu \approx \langle n\rangle + \sigma_n^2/\langle n\rangle \approx 16.3$ (using $\sigma_n \approx 3.5$ measured from our simulations), so $a > \nu$, $b < 0$, and $r > 0$. The inversion is a consequence of the 3D geometry: the AW intercept a exceeds the mean excess degree ν once the topological Euler constraint is lifted.

- *Universality.* All bias levels converge to the same asymptotic degree distribution (Fig. 3), fixing $\langle n\rangle$ and $\langle n^2\rangle$ and hence the linear combination $a\langle n\rangle + b = \langle n^2\rangle$. The individual values of a and b converge because the asymptotic Voronoi geometry is set by seed density alone, not by the division rule.

- *Biased foam more assortative than Poisson.* The division cascade (each event simultaneously increments the degree of an entire neighbourhood)

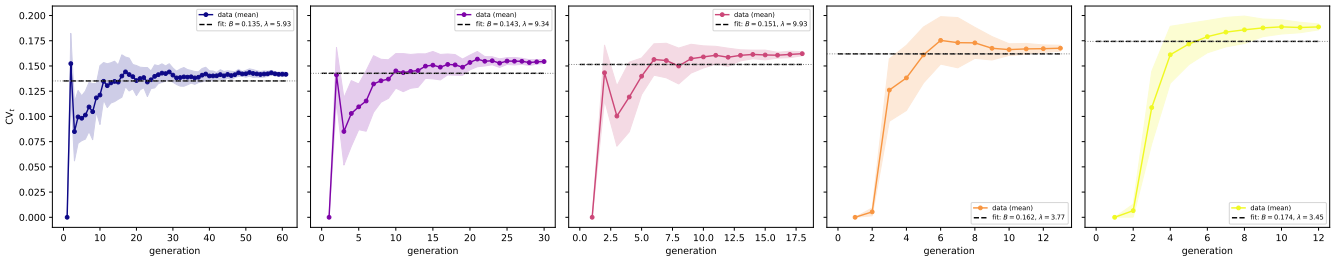

FIG. 10. Exponential decay fits $\mathrm{CV}_t = Ae^{-\lambda t} + B$ for each bias level, with floor B marked. Shaded bands are ± 1 std across 20 runs. The fitted floor is $B \approx 0.15$–0.20 across all bias levels.

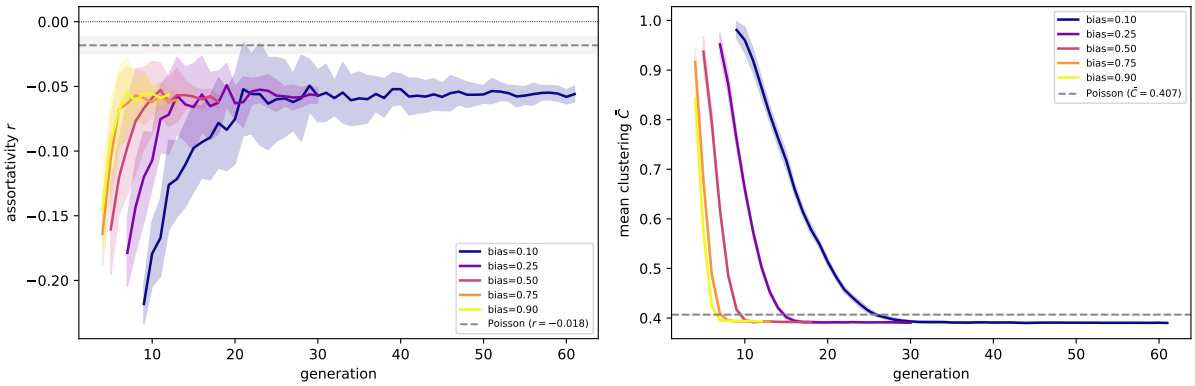

FIG. 11. Newman assortativity r (left) and mean clustering coefficient $\bar{C}$ (right) versus generation. Lines show means over 20 runs; shaded bands are ± 1 std. The gray dashed line marks the Poisson-Voronoi reference. Assortativity passes through zero and converges *above* the Poisson baseline; clustering decays *to* the Poisson baseline. Higher bias reaches the asymptote in fewer generations.

pushes a above the Poisson-Voronoi value, increasing $a-\nu$ and hence r beyond the geometric baseline.

The biased division rule preferentially splits large cells regardless of their face count. A cell's volume therefore reflects how recently it divided, not how many faces it has, driving $\mathrm{Cov}(V,n) \to 0$ in steady state. Since $R^2 = \mathrm{Cov}(V,n)^2/[\mathrm{Var}(V)\,\mathrm{Var}(n)]$, this forces $R^2 \to 0$. The residual volume variance (the CV floor) is then set by the irreducible geometric noise from the stochastic orientation of the division axis: even with perfect preferential division, daughters cannot control their volumes exactly, establishing a hard lower bound on CV. Lewis failure and the CV floor are therefore two manifestations of the same decoupling: the division rule removes the volume-topology correlation while geometric noise prevents full volume equalisation.

B. The Inverted Aboav-Weaire Law

The sign flip of b, and its universality across bias levels, are explained analytically by Eq. (4). Under periodic boundary conditions the $a + b/n$ form fits the full $m(n)$ curve well at maturity, confirming that the inversion is a genuine structural property of the biased foam. Preliminary open-boundary simulations produced a spurious non-monotonic $m(n)$: the curve rose at moderate n then declined at $n \gtrsim 20$. This was a boundary artefact—high-degree cells near the domain wall had neighbours outside the domain excluded from measurement, artificially depressing their counted $m(n)$—and disappears under periodic boundaries.

At the geometric level, the increasing $m(n)$ arises from *spatial locality* of seed density: a cell's face count reflects local seed crowding, and cells sharing a neighbourhood share the same crowded environment, creating positive degree autocorrelation. The biased division process amplifies this through the division cascade: when a cell divides, every neighbour simultaneously gains a face, reinforcing degree–degree correlation.

Equation (3) can be inverted to express a directly in terms of measurable degree moments and the assortativity r:

$$a = \nu + \frac{r\left(\mu_3/\langle n\rangle - \nu^2\right)\langle n\rangle}{\sigma_n^2}. \tag{5}$$

Using our measured values ($r \approx 0.29$, $\langle n\rangle \approx 15.5$, $\sigma_n \approx 3.5$, and μ_3 from our measured degree distribution) Eq. (5) recovers $a \approx 16.5$, consistent with the direct

fit and providing a closed-form consistency relation between the AW intercept and the network statistics. The sign change in b is not an artefact of any particular bias but a structural property of the geometry, emerging once the foam contains $\gtrsim$ 100–200 cells.

C. The Failure of Lewis's Law

Lewis's Law $V(n) \approx k(n-c)$ holds for static Poisson-Voronoi foams where random seed placement creates a natural correlation between cell volume and face count [7]. The biased division process destroys this correlation: as argued above, a cell's volume reflects its division history, not its topology, driving $\mathrm{Cov}(V, n) \to 0$. The $R^2 < 0.20$ across all bias levels extends the Islam & Hassan finding [13] (2D breakdown at strong bias) to the 3D case, where the law is uniformly weak even at 10% bias.

D. Assortativity and the Network-Scale Signature of Division

The sign change in r marks the transition from a topologically constrained small graph (few cells, graph near-complete, $r < 0$ generically) to a geometry-driven large one ($r > 0$ by Eq. (4)). The asymptote $r \approx 0.29 > 0.17$ (Poisson) is explained by the division cascade amplifying $a - \nu$ beyond the Poisson-Voronoi value.

The clustering coefficient's convergence to the Poisson floor confirms that triangles in the adjacency graph are determined by local Voronoi geometry (three cells sharing a common edge) and are insensitive to division history. The distinction is therefore clean: the division rule leaves local structure unchanged while creating an excess of long-range degree correlation, a network-scale imprint invisible to triangle-based statistics.

E. The CV Floor as a Geometric Invariant

The CV floor $B \approx 0.15$–0.20 is the residual volume variance after preferential division has been applied maximally. It is set by the geometric noise from the stochastic division axis: daughters cannot control their volumes exactly, so a floor of irreducible heterogeneity persists regardless of how aggressively the division rule targets large cells. The near-universality of B across a nine-fold range in bias suggests the floor is a property of 3D Voronoi geometry rather than of the particular division rule, and may constitute a fundamental lower bound on volume uniformity achievable by any iterative division process on a Voronoi lattice.

V. CONCLUSION

Three-dimensional biased Voronoi foams behave qualitatively differently from their 2D counterparts under the same division rule. The Aboav-Weaire relation survives the move to 3D but with *inverted* sign: $m(n) = a + b/n$ fits well under periodic boundary conditions with universal parameters ($a \approx 16.5$, $b \approx -33$) independent of bias, and the sign of b is determined by a simple inequality between the AW intercept and the mean excess degree. An open-boundary artefact—boundary cells depressing the apparent $m(n)$ of high-degree interior neighbours—produced a spurious non-monotonic curve in preliminary runs; periodic boundaries resolve this cleanly. The same graph identity yields a closed-form expression for a in terms of the measured assortativity and degree moments. Lewis's Law does not survive: the volume–topology correlation is uniformly weak ($R^2 < 0.20$) across all bias levels and generations, stronger evidence of failure than was seen in the 2D WPSL [13]. The coefficient of variation converges to a floor of CV ≈ 0.15–0.20, a geometric limit intrinsic to iterative Voronoi division in 3D.

At the network level, the foam undergoes a sign change in degree assortativity from $r < 0$ (early generations, topologically constrained) to $r \approx 0.29$ (mature foam, exceeding the Poisson baseline of $r = 0.17$), while the mean clustering coefficient decays to the Poisson floor regardless of bias. The network-level signature of biased division is therefore non-local: it is visible in assortativity—a global property—but not in local clustering.

These results have implications for models of biological tissue growth in three dimensions: classical 2D scaling laws cannot be assumed to hold in 3D, the volume uniformity achievable by preferential division has a hard geometric ceiling, and the topology of the resulting adjacency graph carries a measurable, non-local imprint of the division rule that persists in the mature network.

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DATA AVAILABILITY

The simulation code, processed data, and figure-generation scripts underlying this paper are publicly available at `https://github.com/malcolmwigder/biased-voronoi-foam`.

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